

Mutations in the MAPK/ERK Pathway and Their Impact on Downstream Enzyme Production: A Clinical Review

Abstract

The MAPK/ERK signaling cascade is a critical pathway regulating cell proliferation, differentiation, and survival. Aberrant a

Introduction

Receptor tyrosine kinases (RTKs) such as EGFR and FGFR activate RAS GTPases upon ligand binding. RAS-GTP recruits

Dysregulation of this cascade · whether through constitutive RAS activation, BRAF V600E mutation, or loss of negative reg

Results

Figure 1 illustrates the canonical MAPK/ERK signaling pathway. Sorafenib, a multi-kinase inhibitor, targets RAF and blocks

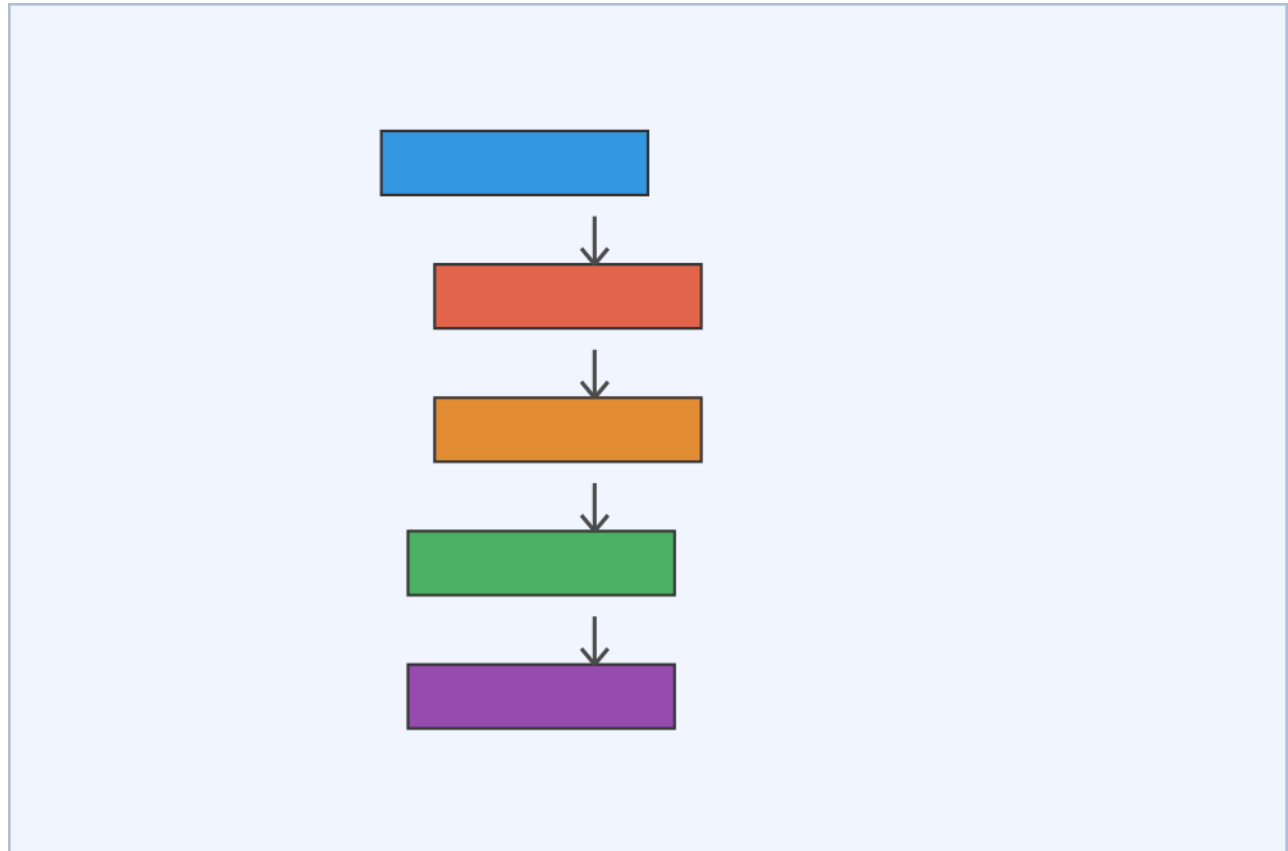


Figure 1: The MAPK/ERK signaling cascade. Growth factor binding activates RAS, which recruits RAF to initiate sequential phosphoryla

Discussion

Our analysis demonstrates that mutations at the RAS and RAF tiers of the MAPK/ERK pathway have a cascading impact on downstream signaling.

Importantly, MEK inhibitors (trametinib, cobimetinib) and ERK inhibitors (ulixertinib) can partially restore normal enzyme levels and signaling.

Conclusions

Gain-of-function mutations in BRAF and KRAS significantly amplify downstream kinase signaling, resulting in aberrant enzyme activity.

References

1. Davies H et al. (2002) Mutations of the BRAF gene in human cancer. *Nature*.
2. Dhillon AS et al. (2007) MAP kinase signalling pathways in cancer. *Oncogene*.
3. Samatar AA, Poulikakos PI (2014) Targeting RAS-ERK signalling in cancer. *Nature Reviews Drug Discovery*.